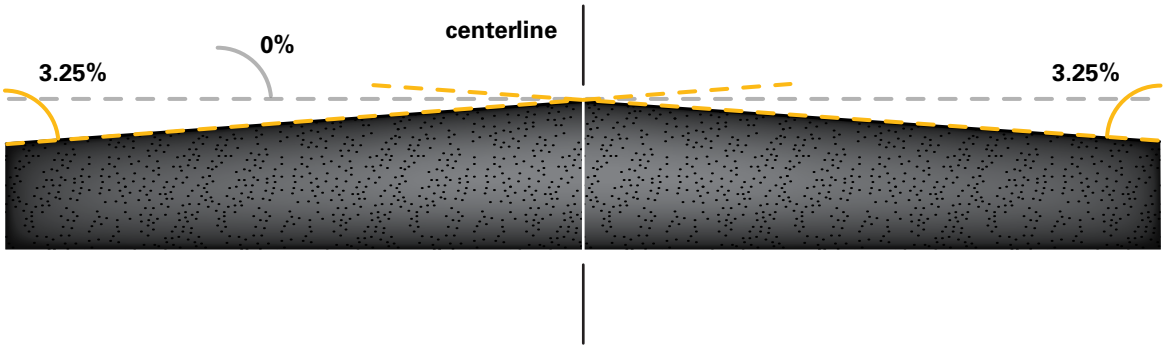
CROWNED STRUCTURE



By crowning the main screed to 3.25 percent and centering the main screed directly over the centerline of the structure, the slopes will be identical on either side of the centerline. To create the desired 4 percent slope right and 2.5 percent slope left, the main screed will require slope. Determining the correct slope calls for some additional calculations.

Subtract the smaller slope from the larger slope:

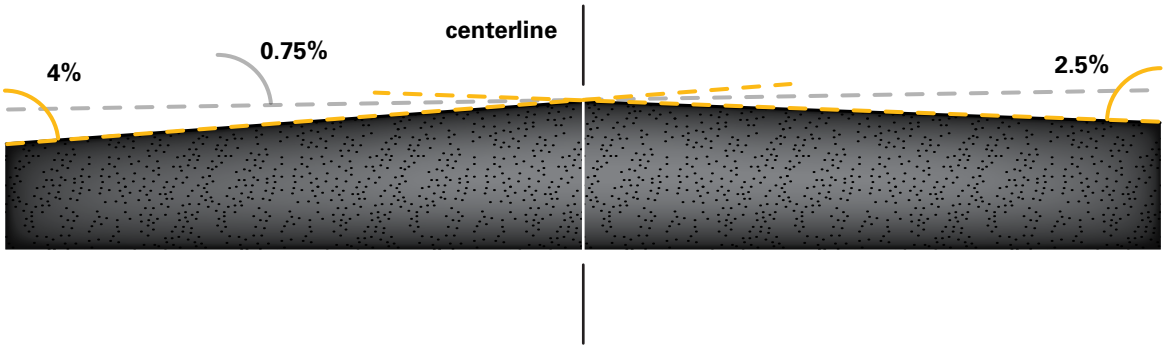
4% - 2.5% = 1.5%

Divide the difference of the slopes by 2:

$$\frac{1.5\%}{2} = 0.75\%$$

The result: 0.75% is the amount of slope.

CROWNED & SLOPED STRUCTURE



The 0.75 percent slope from right to left decreases the right side crowned slope to 2.5 percent, and increases the left side crowned slope to 4.0 percent.

In summary, use slope control when the conditions require corrections to the existing transverse profile or when specified in the project plans. Slope control has two goals. One is to create the proper flow of water toward shoulders and/or drain inlets. The second is to have the proper contours for traffic safety. Be aware that slope control can interfere with yields and ride quality.

[IMPROVING RIDE QUALITY]

Everybody benefits from smooth roads and streets. In addition to the pleasure of operating vehicles on a smooth surface, other reasons are prompting public works departments to focus on specifications that require high levels of smoothness. First, research has proven that smooth roads last longer than rough roads—all other factors being equal. So, life cycle costs decrease; tax dollars have a bigger return on investment; and there is less disruption to traffic flow when repairs or maintenance steps are not as frequent. Second, smooth roads contribute to better fuel economy due to decreased rolling resistance. Less fuel consumed means fewer emissions.

Creating smooth bituminous layers involves managing the four factors that contribute to quality paving. Executing the fundamentals correctly and avoiding big mistakes are important for smoothness. However, the other two factors, paving efficiently and proper use of grade and slope control are even more important.

Most projects that have smoothness specifications lend themselves to efficient paving. Prior to paving, the superintendent or crew supervisor should calculate the paving speed that will minimize paver stops. There are formulas for calculating speed based on hourly tonnage, weight of material, paving depth and paving width. Software such as the Paving Production Calculator simplify the process of calculating paving speeds. (See Unit 2 for details on pre-project planning.)

PAVER SPEED CALCULATOR

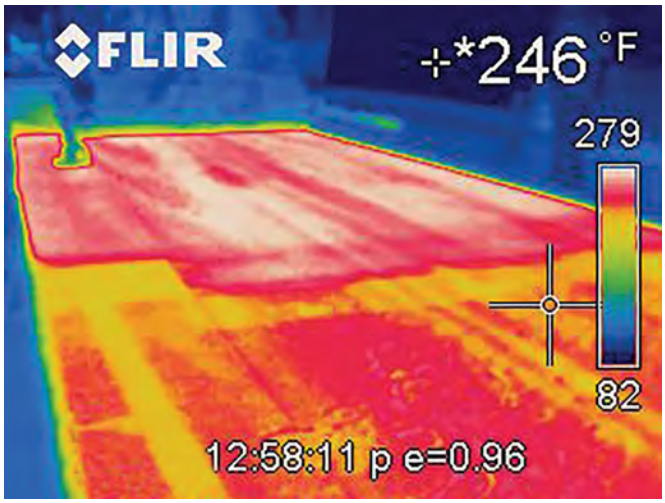
Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal

Exit

General Inputs	*Imperial Units*	*Metric Units*
Paving Thickness:	[2.56] *in*	[65] *mm*
Paving Width:	[16.00] *feet*	[4.877] *meter*
Material Density Uncompacted:	[140] *lbs/ft³*	[2243] *kg/m³*
Paver Speed @ Given Production Rate		
Production Rate of Hot Plant:	[220] *tons/hr*	[200] *tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[15.3] *ft/min*	[4.67] *m/min*
Calculated Paving Speed - 95% Efficiency:	[16.1] *ft/min*	[4.90] *m/min*
Calculated Paving Speed - 90% Efficiency:	[16.8] *ft/min*	[5.14] *m/min*
Calculated Paving Speed - 85% Efficiency:	[17.6] *ft/min*	[5.37] *m/min*
Calculated Paving Speed - 80% Efficiency:	[18.4] *ft/min*	[5.60] *m/min*
Calculated Paving Speed - 75% Efficiency:	[19.1] *ft/min*	[5.84] *m/min*
Effective Paving Speed:	[15.3] *ft/min*	[4.67] *m/min*

Paving speeds vary according to the efficiency factor.

On any project that has long passes, the minimum efficiency factor should be 75 percent. In other words, the paver should be placing material at least 45 minutes out of every hour. Ride quality improves as the efficiency rate goes up. Short interruptions to the paving process, typically during truck exchanges, are often unavoidable and do not necessarily mean that substantial roughness will result. But, the best ride quality occurs when paving is continuous—all other factors being equal. In the previous example, the best ride quality results if the paving speed can be maintained at 4.7 m / minute (15' / minute) for 100 percent efficiency. Ride quality will begin to deteriorate as the efficiency rate goes down. The crew can still place the hourly production rate by adjusting the working speed upward a little, but stops tend to create some roughness.



Temperature differential created by stopping the paver.

Paver stops can create roughness in two ways. One is by creating a temperature differential between the portion of the bituminous layer confined by the screed and the portion of the layer that is directly behind the screed in the area where the initial phase compactor cannot reach. Depending on the ambient temperature and the thickness of the layer, the material may lose heat rather quickly. When the temperature differential exceeds 15° C (59° F), there is likely to be a compaction differential. In other words, the hotter material confined by the screed will compact at a different rate than the colder material exposed to the ambient conditions behind the screed.

User Tip: *To determine how long it will take for a given layer thickness to lose 15° C (30° F) under specific ambient conditions), use PaveCool software. PaveCool is available as a download from the Minnesota Department of Transportation.*



Screed imprint in the bituminous layer created by stopping the paver.

The other way that paver stops may contribute to roughness is by creating a screed settlement mark in the surface of the bituminous layer. The depth of settlement marks is affected by several factors.

Mix Parameters

- Temperature (hotter mixes deform easier)
- Grain size (finer mixes deform easier)
- Gradation curve (well graded mixes deform easier)
- Stiffness (stiff mixes deform less)

Paving Parameters

- Layer thickness (thicker the mat, the deeper the dent)
- Paving width (the more extension, the deeper the dent)
- Ambient temperature (hotter the day, deeper the dent)
- Head of material (lower the head of material, deeper the dent)
- Length of stop (longer the stop, deeper the dent)

Paver Parameters

- Paving speed (higher the speed, greater the settlement)
- Tamper amplitude (higher the amplitude, lower the settlement)
- Tamper speed (higher the speed, lower the settlement)
- Screed set-up (higher the angle of attack, greater the settlement)

The initial phase roller may erase the settlement mark, but roughness is still a concern. For example, the grade and slope control system may react to the screed settlement when the paver resumes operation. Watch the tow-point. If the tow-point makes a significant move (more than 12 mm / 0.5") when the paver resumes operation, there will be enough of a layer deviation to be felt by a vehicle.

Cat pavers are equipped with a screed counterbalance system that reduces the amount of screed settlement during stops. Some Cat pavers also feature a screed lower lock system. Consult the Operation and Maintenance Manual for instructions about the operation of the screed counterbalance and screed lower lock systems.



Material transfer devices, this one equipped with a windrow elevating head, enable the paver to operate continuously at the same speed.

The use of material transfer devices makes high efficiency rates possible. With proper planning, it is possible to pave for hours without stopping or changing the paving speed.

Material transfer devices include machines such as windrow elevators that attach to the front of the paver and are pushed by the paver. Self-propelled material transfer vehicles that operate independently of the paver are also available. Some material transfer vehicles have as much as 23 tonnes (25 tons) of surge capacity. Inserts are often installed in the paver's hopper to increase the amount of material that is available while a new truck is being positioned to supply the transfer device.

It is common to have as much as 36 tonnes (40 tons) of total material available for production between trucks. Do not consume all of the material that is stored in the transfer vehicle or hopper insert when continuing to pave. Using all the material, especially if it is a large-aggregate base material, can cause segregation in the bituminous layer.

Assume that 27 tonnes (30 tons) is available for paving between trucks and a layer that is 65 mm (2.50") thick and 3.66 m (12') wide will be placed. The planned production rate is 200 tonnes per hour (220 tons per hour). The calculated paving speed at 100 percent efficiency under those conditions is 6.25 m / minute (20.5' / minute).

YIELD CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal
Exit

General Inputs	Imperial Units	Metric Units
Paving Thickness:	[2.56] in	[65.0] mm
Paving Width:	[12.00] feet	[3.658] meter
Material Density Uncompacted:	[140] lbs/ft³	[2243] kg/m³
Truck Capacity or Total Tonnage:	[30.0] ton	[27.2] tonnes
Length of Mat at 100% Yield:	[167.41] feet	[51] meter
Actual Length of Mat produced:	[0.00] feet	[0.00] meter
% Yield for given truck load or tonnage:		[0]

Thickness:	[2.56] in	[65.0] mm
Length of Mat produced:	[0.00] feet	[0.00] meters
Width:	[12] feet	[3.658] meters

The yield calculator provides length of paving for a given weight of material.

Use the Yield Calculator function of the Paving Production Calculator. If 27 tonnes (30 tons) are available in the transfer vehicle and insert, a length of 51 m (167') can be paved. At a paving speed of 6.25 m / minute (20.5' / minute), there will be just over eight minutes of paving production. If it is taking more than eight minutes to complete a truck exchange, something is wrong with the procedure. A truck exchange should take no longer than three minutes unless severe traffic congestion is encountered.

Caterpillar recommends the use of material transfer devices on projects that have a ride quality specification or anytime there is a need to significantly improve smoothness. Non-stop paving at a continuous speed will produce a 10-15 percent smoother ride compared to stop-and-go paving—all other factors being equal.

In addition to continuous paving at a constant speed, installation of averaging skis is another proven way to create better ride quality.

Mechanical averaging skis are suitable for applications where there are long passes and few obstacles such as drain inlets and utility boxes. Mechanical skis like the Cat 9-meter (30') Outboard Leveler should be installed when there is a suitable grade reference outside the paving width. Mechanical skis like the Cat Fore 'N Aft Leveler should be installed when the only suitable grade reference is inside the paving width. Caterpillar recommends using mechanical averaging skis on both sides of the paver when there is moderate to severe roughness in the surface to be paved.



Dual, mechanical averaging skis can produce up to a 60-70 percent improvement in one-pass smoothness.

Granular bases often have a high degree of roughness. Some milled surfaces have moderate roughness as well. Using the International Roughness Index as a reference, moderate roughness can be classified as total deviations between 1.9 m / kilometer and 2.8 m / kilometer (120" / mile and 180" / mile). Extreme roughness as measured by the International Roughness Index would be deviations in excess of 2.8 m / kilometer (180" / mile).

Caterpillar research indicates that a paver equipped with mechanical averaging skis on both sides and paving at a continuous speed can reduce roughness by 60 – 70 percent in one pass when moderate to severe roughness exists. For example if the grade deviations total 2.3 m / kilometer (150" / mile) prior to paving, the predicted deviations will be lowered to around 80 cm / kilometer (53" / mile) in one pass when dual mechanical averaging skis are installed on the paver.

During installation of another bituminous layer using a paver equipped with two mechanical averaging skis, another 20-40 percent smoothness improvement can be realized. Using the previous example, expected deviations can be lowered to around 56 cm / kilometer (37" / mile). The percentage of smoothness improvement will be even less if a third layer of bituminous material is added.

Non-contact skis are more versatile and more convenient than mechanical skis. Non-contact skis can be installed to reference grade inside the paving width or outside the paving width. Non-contact skis use electronic averaging to reduce roughness. The roughness reduction factor of non-contact skis is not quite as high as the roughness reduction factor of mechanical averaging skis. Non-contact averaging skis are most suitable on projects where there is light to moderate roughness in the surface to be paved. Excellent smoothness results have been obtained with non-contact skis on projects where the milled surface has the correct transverse profile and a reasonably smooth longitudinal profile.

Non-contact skis do not have to be removed in order to move the paver around the worksite. Therefore, Caterpillar recommends installing non-contact skis on projects with time constraints, like night shifts. Non-contact skis are not affected by obstacles like drain inlets and are preferred for their convenience by many crews.



Non-contact skis offer convenience and adequate averaging for surfaces with light to moderate roughness.

Another technique that can help improve smoothness in driving lanes is to pave the emergency lane (shoulder) first. When the emergency lane is paved first, it becomes an improved grade reference for the averaging ski when the paver works on the adjacent driving lane. Plans don't always specify paving the emergency lane first, but it is a good idea to ask for permission.



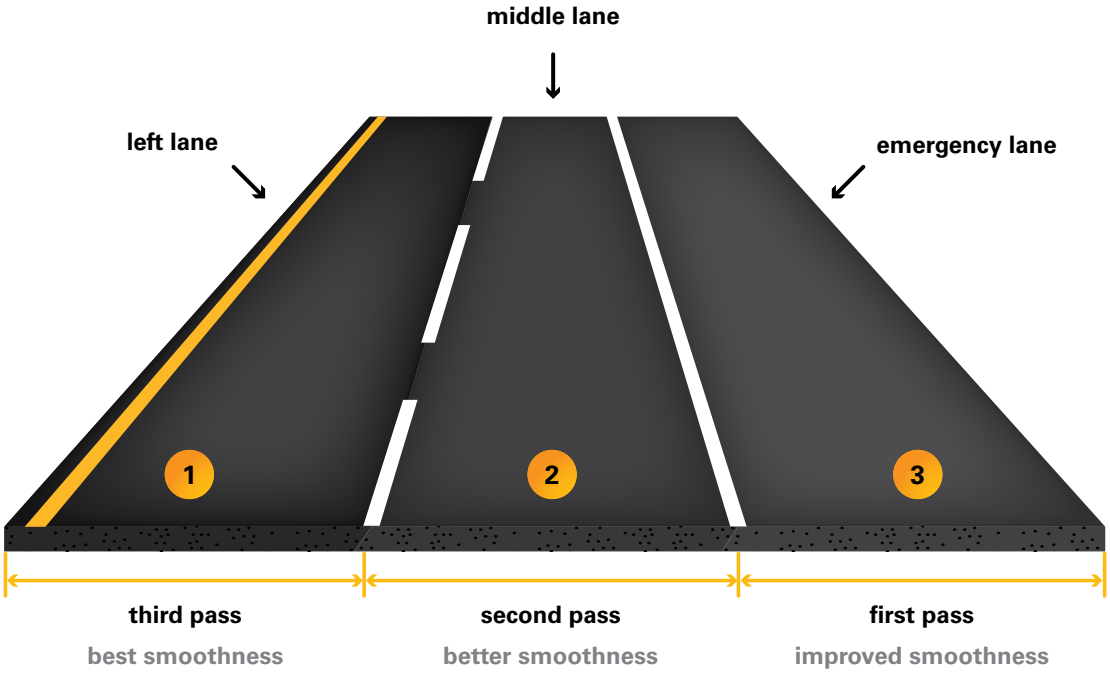
Paving emergency lanes first improves the smoothness of adjacent driving lanes.

The emergency lane that has been paved first should be a better grade reference than a milled surface, or granular base or even an existing bituminous surface. Plus, the emergency lane provides a place for the compactors to stop and reverse without leaving marks in the new layer of the driving lane. Remember, the emergency lane is not measured for smoothness. Only driving lanes are measured.

Smoothness will continue to improve as the paver completes adjacent driving lanes because the smoothness of the grade reference is constantly being improved.



Averaging ski using emergency lane as a grade reference.



Smoothness of the emergency lane should be improved when compared to a milled surface, for example. Then, the smoothness of the driving lane adjacent to the emergency lane will be better than the score of the emergency lane. Finally, the far left driving lane will have the best smoothness.

In summary, to maximize the smoothness of a bituminous layer, plan for continuous paving at a constant speed. Pave with averaging skis installed on both sides of the paver whenever possible. Choose the type of averaging ski that is appropriate for the application. Avoid the use of slope control unless it is a requirement of the specification or if the transverse profile needs correction. Whenever possible, pave the emergency lane first to provide an improved grade reference for the driving lanes.

[PAVING TO ELEVATIONS]

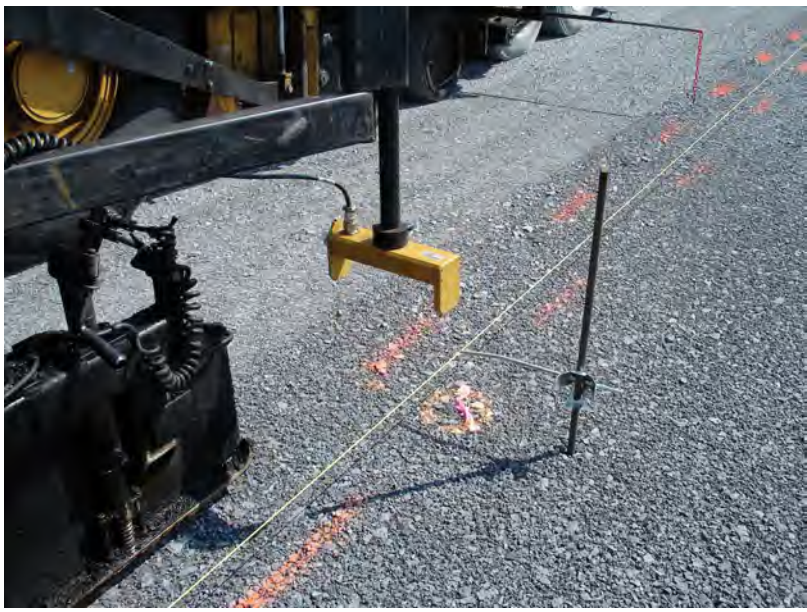
As it relates to asphalt paving, the term elevation means the height above sea level of any geographic location. Various types of survey equipment, sometimes in conjunction with Global Positioning Satellites, measure elevations on construction projects. Surveyors record elevations, and engineers use them to help create the plans for a project. Some construction equipment that uses three-dimensional systems mentioned in Unit 6 incorporate the engineering plans in their guidance systems. More commonly, at least at the time of this guidebook's publication, the elevations of a bituminous structure, such as a highway or airport runway, are controlled by referencing a stringline or grid marks on a surface.



Controlling base layer elevation with an erected stringline on new highway construction.

User Tip: *The Cat Grade Control System has a feature called "cross coupling." When cross coupling is left at the default setting of 100 percent, slope control acts simultaneously with grade control corrections. There is no delay in tow-point movement on the slope control side. Slopes will be more accurate and the crew will have an easier time with joint matching as they pave any adjacent lane.*

A stringline can be used as the reference for elevation. The stringline can be a temporary device used for one phase of construction. Or, a stringline can be used for many phases of earthwork and paving all the way to the final bituminous surface layer. The stringline is a reference for one side of the paver. A sonic grade sensor or a mechanical grade sensor with a wand follower can be used to reference a stringline. The other side of the paver can use grade control or slope control. The use of slope control, either manual or automatic, is more commonly used on the side opposite the stringline. The grade side assures that the elevations are correct while the slope assures that the transverse profile will direct water in the right direction. Additionally, the elevations will transfer correctly across the width of the layer as long as the slope is correct.



Cat Grade Sensor in stringline orientation.

There are a few things to remember when using the Cat Grade Sensor. First, orient the sensor so it is perpendicular to the stringline with the cable connection on the left side of the sensor when facing forward. Position the sonic sensor 40 to 45 cm (16 - 18") above the string and centered as much as possible over the stringline. The three center transducers will read the stringline and the average of those three measurements provide the signal sent to the control module. The outer two transducers enable horizontal guidance for the display.

When the operator selects the Sonic Sensor over Stringline icon in the Machine System area, the position of the sensor over the stringline is shown in the Guidance area. Five guidance symbols provide excellent visual reference.

A green circle around the yellow string means the sensor is centered over the stringline. (As shown in the adjacent illustration.)

A red arrow on either side of the yellow string means the sensor is slightly off center, left or right, but still functional and accurate.

If the center of the sonic sensor moves too far away off the center of the stringline, then the red arrow remains and the entire guidance symbol will be highlighted. As long as the stringline is more than 5 cm (2") above the grade or body of the ski, the sensor will lock to grade. Grade control is no longer functional and will not operate until the operator re-centers the sensor over the stringline.



A stringline guidance symbol will appear on the screen when Sonic Sensor over Stringline has been selected. This view shows that the sensor is centered over the stringline.

Sometimes a stringline is not erected. Rather, elevations are marked on the grade to be paved. The elevations are laid out in a grid with marks spaced every 7.6 m (25'). The screed operator looks ahead to determine what elevation difference, if any, exists at the next mark.



A survey crew may mark elevations directly on the grade in a grid pattern.

The screed operator wants the elevation change, if any, to occur in less than 7.6 m (25'). Assume, for example, that the elevation change is 6 mm (0.25") upward from one survey mark to the next. The operator will push the UP arrow on the display five times. Grade control changes occur in increments of 1.3 mm (0.05") for every time the UP or DOWN arrow is pushed.

With the grade sensor located just ahead of the auger chamber, tow-point movement will be about three times farther than the distance of the correction that was called for. In this case, the tow-point will move up about 18 mm (0.75"). The screed will begin to move up immediately and will complete its movement to a new elevation in about 3.5 m (12'). A sensor located just ahead of the auger chamber is a good choice for following elevation changes that are 7.6 meters (25') apart.



Grade sensor position just ahead of the auger chamber helps create fast screed reaction.

The speed of the screed reaction can be affected by adjusting the Elevation Valve Speed. Enter the Installation Menu on the Home Display. Select Control Settings, then Valve Speeds, and finally Elevations. The default value is 50 percent. If faster screed reaction is needed, use the slider on the touchscreen to increase the value of the valve speed. Adjust upward, for example, to 75 percent and verify that elevation changes are occurring within the grid distance.



Increasing the value of the Elevation Valve Speed also speeds up screed reaction.

User Tip: *Caterpillar recommends using automatic grade control when paving to meet exact elevations. Grade control is extremely accurate and can be made to act quickly. If manual elevation control is used, the change in elevation will occur over a distance of about five tow arm lengths or 12 meters (40').*



Producing a straight edge to match is the first step in longitudinal joint construction.

[CONSTRUCTION OF LONGITUDINAL JOINTS]

Poor longitudinal joint construction is recognized as one of the major problems in bituminous pavement construction. Considerable research has been done over the years with no universal solution other than eliminating the joint. The solution to the problem, as dictated by some public works departments, is to pave wider passes with no joints between lanes or to pave in echelon. Realistically, these solutions are not always possible and cost effective. Echelon paving and full-width paving techniques, covered later in this unit, may require closing entire roads for paving and possibly stretching capacities for pavers, rollers, and asphalt plants. This section assumes longitudinal joint construction consisting of a hot / cold or hot / warm joint.

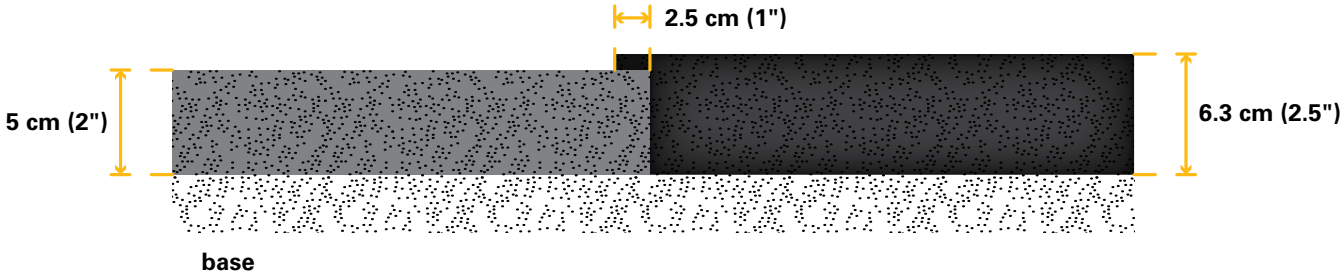
In order to be successful at longitudinal joint matching, the crew has to have an unconfined edge that is straight and reasonably smooth. Prior to the start of paving, create a steering guide for the paver operator. Lay out the paving pass and position the paver and screed extensions so any changes to the paving width are accomplished using the side of the paver that will not have a joint match. Caterpillar recommends running the end gate skis down so they float on the grade being paved and create a straight vertical face on the edge that will be matched.

To promote the bond between the cold side of the joint and the hot side, the face of the joint should be thoroughly covered with tack or joint adhesive. Be careful not to overlap the joint with the spray bar too much onto the cold side. The end gate ski will ride on this surface and may be affected by the sticky tack. Plus, the area close to the joint may be used as reference for a mechanical, contact-type averaging ski. A contact ski can be affected by the build-up of tack on its runners or shoes.



Tack or adhesive must be applied to the face of the joint, as well as the base prior to paving.

CORRECT SQUARE JOINT — END GATE DOWN



With the end gate ski floating on the cold side, overlap the cold side up to 25 mm (1").

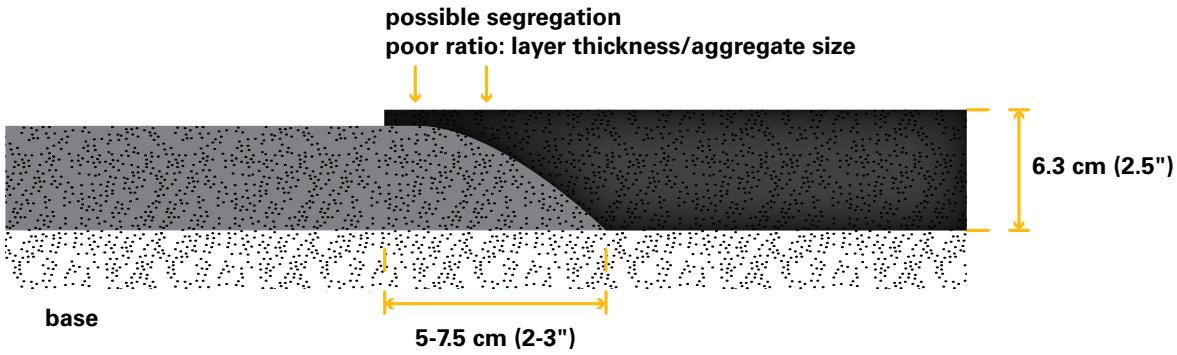
Once paving has started, maintain a slight overlap onto the cold side of the joint. When matching a straight line, the extender should not have to constantly move in and out. Up to 2.5 cm (1") overlap is acceptable. If there is excessive overlap, a laborer will have to rake the extra material away from the joint over onto the hot side of the joint. Maintain the height of the hot mat according to the compaction rate of the material. For example, to match a compacted layer that is 5 cm (2") thick, the thickness of the hot matching layer may be 6.3 cm (2.5"). If there is an incorrect height match, a laborer will have to rake the joint.

While light bumping of the joint is acceptable, longitudinal joint raking should be avoided as much as possible. Raking tends to push material away from the face of the joint and, depending on the size of the aggregate in the bituminous mixture, can cause a segregation stripe adjacent to the joint. Raking has a negative effect on joint durability as it increases permeability allowing moisture penetration. The goal should be to always construct longitudinal joints that do not need to be raked.



Too much overlap or incorrect height match can cause excessive raking of the joint.

INCORRECT SQUARE JOINT — END GATE UP



Running end gates up creates joint problems.

When creating an edge for joint match, always run the end gates down on the grade. When paving with the end gates up, which some operators like to do to gauge layer thickness, the edge of the layer will be rounded, especially after compaction. The overlap is too much when the joint match pass extends on to the cold side. And, more importantly, there will be an area over the joint where the screed is dragging aggregates due to loss of layer thickness. The compaction process will fracture aggregates in this area. The two flaws that should be avoided at the longitudinal joint—poor density and high permeability—will be more likely to occur.

Specialty longitudinal joints may be required. Some public works departments specify notched wedge joints for two reasons. First, in some high traffic areas, it is prohibited to have an unconfined edge open to traffic if the edge is 5 cm (2") or higher. Second, if the notched wedge is correctly constructed, the longitudinal joint density tends to be as high as conventional vertical edge joints.

The notched wedge attachment fits inside the screed extension. The height of the notch and the angle of the wedge are adjustable.

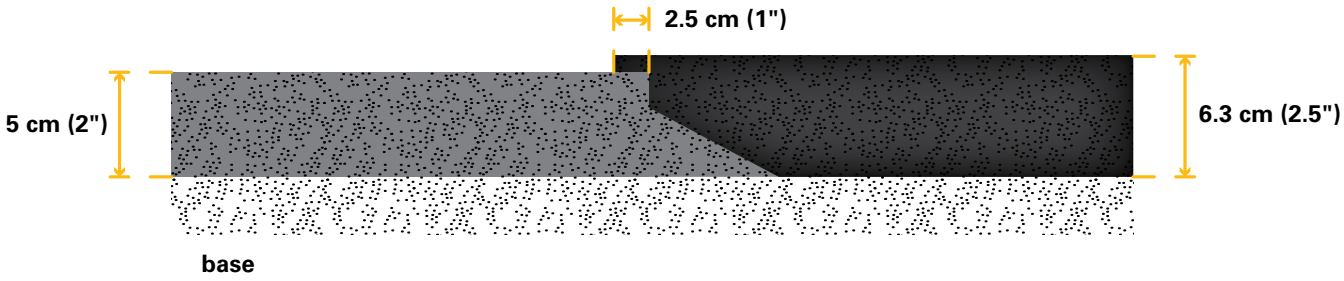
Some crews tow a small steel drum (shown to right) or rubber tire behind the screed to compact the wedge. Other crews use a tamper plate compactor to compact the wedge.



Notched wedge joints are required on some projects.

A notched wedge joint should overlap the same way that a vertical face joint would overlap. Keep the overlap at 25 mm (1") maximum and calculate pre-compaction height of the hot layer.

CORRECT NOTCH WEDGE



Overlap a notched wedge joint the same as a vertical face joint.

User Tip: *One very important consideration about notched wedge joints is the height of the notch. The height of the notch should be 1.5 times greater than the size of the largest aggregate in the bituminous mixture. This height clearance will allow mix to flow through the notch area without dragging. If drag marks are visible at the intersection of the hot and cold layers, increase the depth of the notch.*

Caterpillar recommends using automatic grade control for matching the height of the cold layer. Matching the height of the cold layer can be accomplished manually, but automatic grade control is easier and more accurate.

At one time, most crews installed a single sensor on the joint match side. The sensor would usually be held by mounting hardware attached to the screed end gate. The sensor would always be relatively close to the auger chamber, a sensor position that creates a fast-acting screed.

Cat Grade Control enables the crew to install mounting hardware on the screed end gate or on the tow arm. For the most versatility, Caterpillar recommends installing grade sensor mounting hardware on the tow arm. There is one installation position toward the rear of the tow arm. There is another installation position in the middle of the tow arm. A third position is just behind the tow-point cylinder at the front of the tow arm. The mounting arm pivots so the sensor can be positioned to match the application requirement for screed reaction.



A single sensor is often used for height match on a longitudinal joint in parking lots and streets.

As long as the compacted layer is reasonably smooth, the screed does not have to be extremely reactive. Crews will often position a grade sensor about halfway between the auger chamber and the tow-point connection. (See photo on page 216.) The screed will be responsive, but will not make large, immediate height changes with a single grade sensor at the mid-point of the tow arm.

If a very fast screed reaction is desired, leave the sensor mounting hardware in the same position and simply pivot the sensor back toward the auger chamber.

A single grade sensor set-up is suitable for joint matching in street and parking lot applications where there is no specification for ride quality.

User Tip: *Remember, it does not matter where the mounting arm is attached. What affects screed reaction is the position of the grade sensor relative to the tow-point connection or the auger shaft. Many crews have been trained to install the mounting hardware on the screed end gate. Installing the mounting hardware on the end gate limits the versatility of the grade control system. Use the tapped holes in the tow arm to attach the mounting hardware.*

Increased utilization of ride quality specifications has prompted many crews to perform joint matching with averaging skis. At one time, crews thought that an averaging ski could not provide adequate height match because the screed moves very slowly when the grade sensor uses an averaging ski for grade reference or when an electronic averaging ski is installed.

If the compacted layer does have a moderate to severe level of roughness, then the height match provided by an averaging ski will not be accurate. There will be some high spots and some low spots at the longitudinal joint. In that case, it would be better to use a single grade sensor for improved height match. However, if the compacted layer was placed under the control of one or two averaging skis, its ride quality will normally be acceptable and an averaging ski will provide an acceptable height match with the adjacent layer.



Joint matching with averaging skis is common when there is a ride quality specification and the adjacent layer is reasonably smooth.



Pre-heating the cold, compacted layer is specified on some applications.

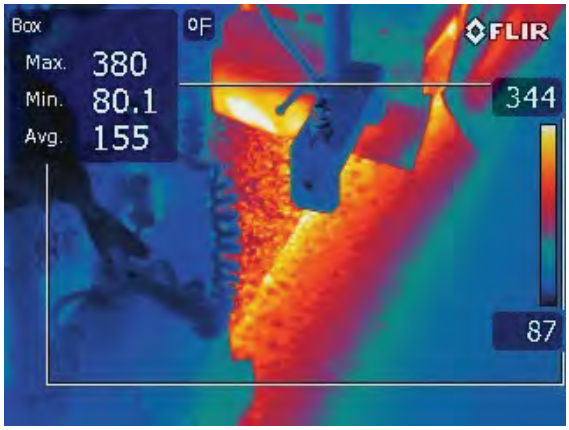
A longitudinal joint that is properly constructed, treated with a good adhesive, and properly compacted should resist moisture penetration and stay bound together. Another technique that can help longitudinal joints last longer is pre-heating prior to compaction.

Research conducted by public works agencies, associations and universities has proven that pre-heating the cold, compacted layer just prior to joint matching and compaction increases joint density and decreases permeability. For some applications, joint heating is a method specification mandated by the project owner.

There are various types of joint heaters. Some attach to the paver and are towed by the paver. Others are towed by some other vehicle. Often, the decision to tow with the paver or another vehicle is determined by the width of paving.

Joint heaters generate high temperatures, but not so high as to damage the properties of the asphalt cement used as binder. A propane heater with infrared technology can heat the cold joint to a typical average of about 120° C (250° F) while being towed at normal paving speeds. The cold layer becomes flexible at the joint to a depth of around 25 mm (1") for improved bonding with the adjacent hot layer.

In summary, the construction of longitudinal joints is one of the most important responsibilities for the paving crew. The main elements of good joint construction are: straight line easy to match; correct overlap up to 25 mm (1"); correct height allowing for compaction rate; and no raking.

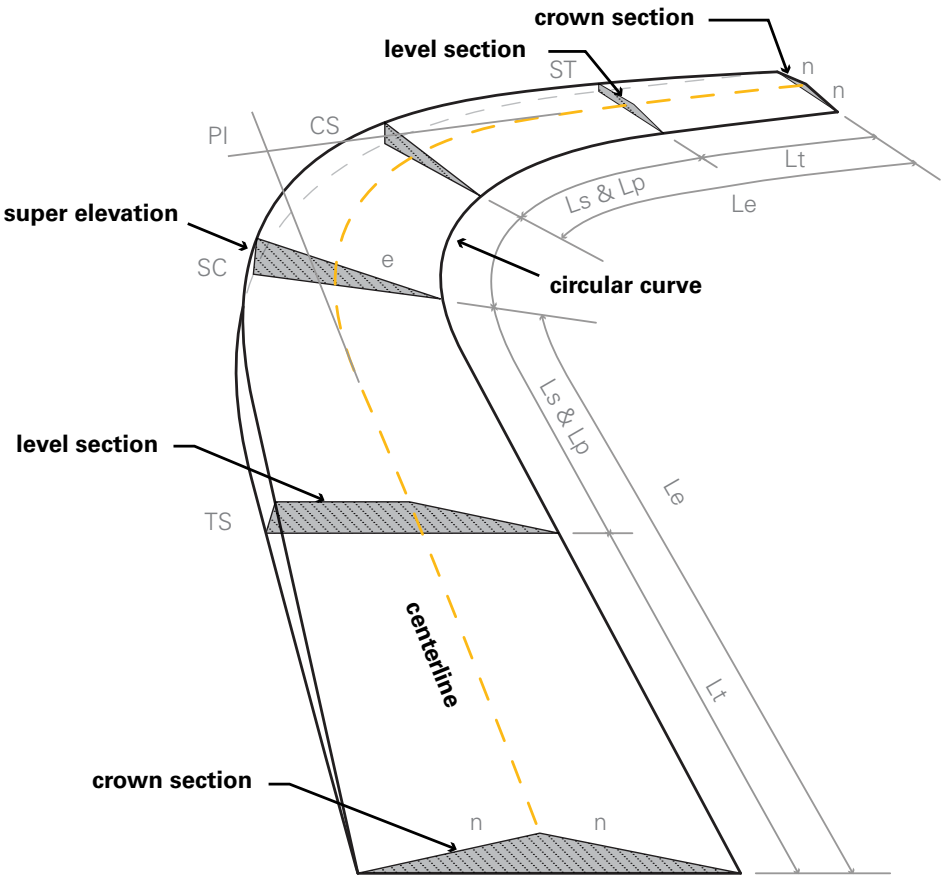


Heat from the joint heater penetrates the cold layer at the longitudinal joint.



[PAVING SUPERELEVATIONS]

The curved sections of highways and streets are commonly constructed with transverse profiles called superelevations. In the superelevation, one edge of the roadway is elevated higher than normal compared to the other edge. This superlevation is classified in terms of crossfall, or slope, rather than in centimeters or inches. The higher than normal slope creates a "banked" section in the curve that helps counteract the centripetal acceleration produced as a vehicle rounds the curve. The normal maximum slope for superelevations in high-speed highways is 6 percent. Higher slopes may be used for lower traffic, lower speed roads and streets.



Typical Superelevation Diagram

Being familiar with the sections that make up a typical super elevation is important. At the entrance to the curve, the structure has a crown at the center. That is called the crown section. In this example, there is a 2 percent slope on either side of the crown in the center of the roadway.

At some distance in front of the crown section is the level section. At the point of the level section, the upper half of the structure (left lane) is flat, 0 percent slope. The lower half of the structure (right lane) still provides a 2 percent slope. The distance between the crown section and the level section is referred to as the tangent runoff. The transition between the crown section and the level section (tangent runoff) should occur as a gradual change in the transverse profile.

At some distance in front of the level section is the superelevation. This is the point of the structure's maximum slope. There is no crown in the structure. In this case, the slope is 6 percent from left to right. The distance between the level section and the superelevation is referred to as the superelevation runoff. Again, the transition between the level section and the superelevation should be gradual.

The distance between the normal pavement section and the maximum slope of the superelevation is the distance that is used to develop the maximum slope in the curve. That distance varies depending on the type of roadway and speed limits.

The profile of the superelevation will be constant at the maximum slope for some distance in the middle of the curve.

As the curve begins to straighten, the transverse profile begins to change back to the level section where half of the structure is flat and half has 2 percent slope.

Finally, at some distance where the roadway is straight, the normal crown section with 2 percent slopes begins again.

From the standpoint of the paving crew, how they set up to pave the superelevation depends on the quality of the work done in front of them by the base preparation crew (new construction) or by the milling crew (cold plane and repave) or by the condition of the existing structure (overlay).

Prior to paving, check the slopes of the grade to be paved at various locations for the length of the curve. Use a 2.5 m (8') slope board or a calibrated electronic slope meter.

The slopes should be clearly marked on stakes adjacent to the roadway. Prior to paving make sure that all the stakes are in place and easy to see from a distance of at least 15 m (50'). Check the slope of the grade at each stake location and between the stakes as well.



Check slopes throughout the length of the superelevated curve.



Grade stake showing slope at a point in the superelevation runoff.

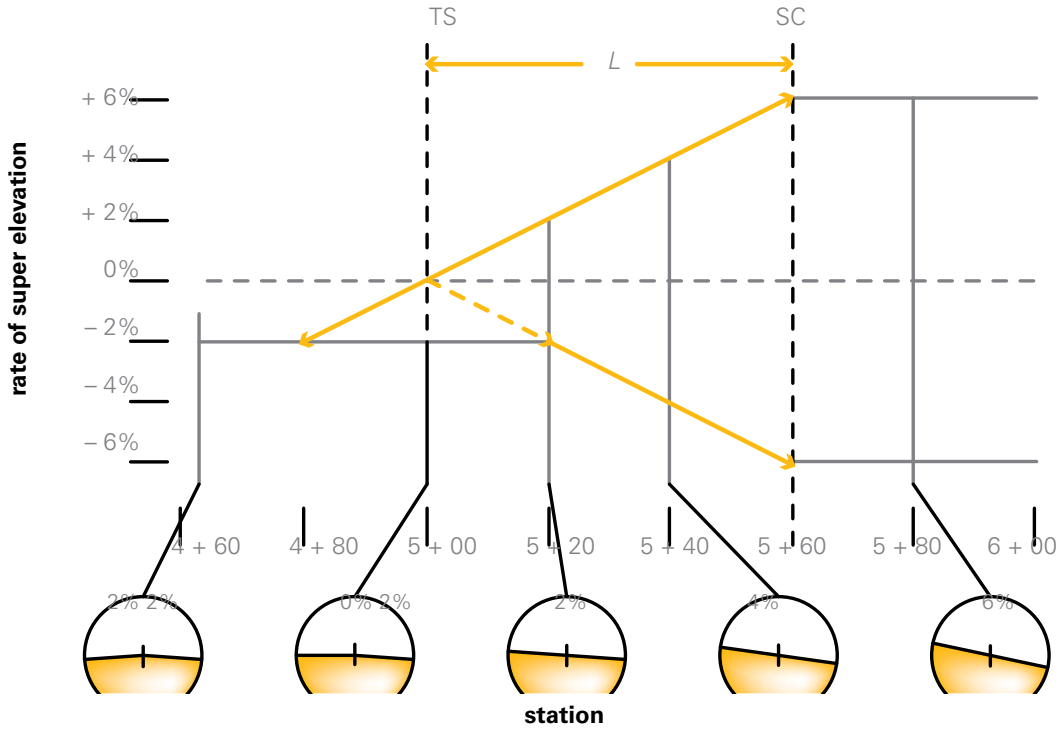
If the slopes are all within 0.2 percent of the slopes that are marked on the grade stakes, grade control can be used to pave the superelevation. If the slopes that are close (within 0.3 percent to 1 percent) to the slopes marked on the grade stakes, automatic slope control can be used to make precise slope changes that match the design. If the slopes reveal that corrections in excess of 1 percent need to be made, then manual slope control should be used.

Remember the formula used to calculate elevation changes when using slope control. For every 1 percent of slope correction, the elevation (layer thickness) will change 1 cm per meter (0.125" per foot) of paving width. When using automatic slope control to make large slope corrections, it is likely that the tow-point cylinder will run out of stroke. The slopes will not be accurate and the operator will be forced to "gain back" tow-point cylinder while trying to pave the superelevation. (Refer to the section, Paving with Slope Control, in unit 6 on pages 167-168.)



Electronic slope meter showing 3.6 percent slope from right to left adjacent to where the grade stake on page 221 shows 4 percent required.

Cross sections show the profile of the roadway at various stations on a project. Thinking of the cross sections as two separate paving passes makes it relatively easy to understand how to set up and control the screed and create the profile for one side of the structure.



Superelevation diagram, showing roadway cross sections.

Assume that paving of the structure starts on the right side of the centerline. At the crown section, the screed has no crown and there is a 2 percent slope from left to right. At the level section and at station 5 + 20, there is still 2 percent slope from left to right. Slope begins to increase gradually until it reaches 6 percent at station 5 + 80. Slope eventually reverses this sequence until the end of the curve and a new crown section starts.

Now, assume that paving is occurring on the left side of the structure. This layer will be a joint match to the right lane. At the crown section, the screed will have a 2 percent slope from right to left. Begin to decrease slope until 0 percent is reached at the level station, 5 + 00. Then following marks on the grade stakes begin to create a left to right slope and gradually climb to 6 percent slope by the time station 5 + 00 is reached. Then, reverse the sequence to start a new crown section at the end of the curve.

Now assume that full width paving will occur through the superelevation. Before starting, make some calculations and use them as a reference when paving the superelevation.

STATION 4 + 60, CROWN SECTION

The crown section is the normal profile when paving full width. The center of the main screed is aligned with the centerline of the structure. The main screed has a 2 percent crown.

STATION 5 + 00, LEVEL SECTION

Step 1: Sum of the slopes divided by 2 = crown

$$\frac{0\% + 2\%}{2} = 1\% \text{ crown}$$

Step 2:

Subtract the smaller slope from the larger slope:

$$2\% - 0\% = 2\%$$

Divide the difference of the slopes by 2:

$$\frac{2\%}{2} = 1\%$$

The result: 1%, is the amount of slope.

Starting at the crown section 4 + 60, begin to gradually flatten the main screed crown to create 1 percent crown at level station 5 + 00. Starting at the crown section 4 + 60, begin to create left-to-right slope ending at 1 percent slope at the level section.

STATION 5 + 20, TRANSITION STATION

The screed is flat at station 5 + 20. In other words, there is one slope running through the centerline. Starting at the level section 5 + 00, begin to gradually flatten the main screed crown to create 0 percent crown at transition station 5 + 20. Starting at the level section 5 + 00, begin to increase left-to-right slope ending at 2 percent slope at station 5 + 20.

STATION 5 + 80, SUPERELEVATION

At transition station 5 + 20, continue to increase left-to-right slope. At station 5 + 40, the slope should be 4 percent. Finally at station 5 + 80, the slope should be 6 percent. Continue to pave at 6 percent until entering the transition out of the curve. At that point, reverse the steps to eventually return to the normal profile of 2 percent main screed crown.

When paving a superelevation using automatic slope control, slowly make the required slope changes. Slope changes are made in 0.1-percent increments using the UP or DOWN arrows on the display or using the grade adjustment knob on some other systems. For example, if it's required to change slope from 2 percent to 3 percent over a distance of 15 m (50'), start to make the changes when the screed is adjacent to the grade stake marked 2 percent.

Try to space out the slope changes so the last change will occur just before the grade stake marked 3 percent is reached. Increase the slope to 2.1 percent. When paving at a speed of 5 m / min (17' / min), it will take about three minutes to reach a grade stake that is 15 m (50') away. Wait about 15 seconds and increase the slope to 2.2 percent. Continue this pattern until 10 slope adjustments of 0.1 percent have been made. This will create a gradual change in slope and the slope should be correct—3 percent when the grade stake marked 3 percent is reached. Remember, slope changes occur quickly. It only takes approximately one meter (a yard) for the change to fully occur. Monitor the tow-point connection. Don't let the tow-point cylinder run out of stroke.



Use the display or control box to slowly make slope changes.



Paving superelevation with averaging ski on left, slope right.

At least one side of the paver will be under grade control while paving through a superelevation. On a highway project, the grade side may be using an averaging ski. Depending on the length of the superelevation and how quickly the slopes change, use of the averaging ski may need to be discontinued. An averaging ski is normally at least 9 m (30') long. The front of the ski may be referencing a significantly different elevation than the rear of the ski in the transition areas of the superelevation and, consequently, the screed reaction may be erratic. If the tow-point starts to make large and frequent moves when entering a transition area in a superelevation, turn the grade control system to Manual mode. Control the thickness of the layer manually until paving of the normal crown section resumes.

In summary, always verify the accuracy of the actual slopes in the superelevation before paving. Be sure that the slopes are clearly marked on grade stakes and easily visible from the screed at a distance of up to 15 m (50'). Depending on the accuracy of the slopes, decide whether to pave with dual grade control, one side grade control and one side automatic slope control, or one side grade control and one side manual slope control.